

y-Sweep

$$\begin{aligned} d_1 &= H - a^T v & d_2 &= H + a^T v & d_3 &= v - a & d_4 &= -v + a & d_5 &= \frac{1}{2} \tilde{v}^T \tilde{v} & d_6 &= d_1 d_4 - d_2 d_3 + d_3 d_6 - d_4 d_5 - d_1 v + d_2 v \\ d_7 &= w_4 - u w_2 + w_1^2 w_1 \end{aligned}$$

$$\begin{aligned} a_1 &= \frac{-1}{d_6} [d_2 w_3 - d_4 w_4 - d_5 w_3 + v w_4 - d_4 u^2 w_1 + w_1^2 v w_1 + d_4 d_5 w_1 + d_4 u w_2 - d_2 v w_1 - u v w_2] \\ &= \frac{-1}{d_6} [w_1 (d_4 d_5 - d_2 v) + w_3 (d_2 - d_5) + w_4 (v - d_4) + u w_2 (d_4 - v) + u^2 w_1 (v - d_4)] \\ &= \frac{-1}{d_6} [w_1 (d_4 d_5 - d_2 v) + w_3 (d_2 - d_5) + d_4 (v - d_4)] \end{aligned}$$

$$\begin{aligned} a_2 &= \frac{-1}{d_6} [d_1 w_3 - d_2 w_3 - d_3 w_4 + d_4 w_4 - d_3 u^2 w_1 + d_4 u^2 w_1 - d_1 d_4 w_1 + d_2 d_5 w_1 + d_3 u w_2 - d_4 u w_2] \\ &= \frac{-1}{d_6} [w_1 (d_2 d_3 - d_1 d_4) + w_3 (d_1 - d_2) + w_4 (d_4 - d_3) + u w_2 (d_3 - d_4) + u^2 w_1 (d_4 - d_3)] \\ &= \frac{-1}{d_6} [w_1 (d_2 d_3 - d_1 d_4) + w_3 (d_1 - d_2) + c_2 (d_4 - d_3)] \end{aligned}$$

$$a_3 = w_2 - u w_1$$

$$\begin{aligned} a_4 &= \frac{-1}{d_6} [d_1 w_3 - d_3 w_4 - d_5 w_3 + v w_4 - d_3 u^2 w_1 + w_1^2 v w_1 + d_3 d_5 w_1 + d_3 u w_2 - d_1 v w_1 - u v w_2] \\ &= \frac{-1}{d_6} [w_1 (d_3 d_5 - d_1 v) + w_3 (d_1 - d_5) + w_4 (v - d_3) + u w_2 (d_3 - v) + u^2 w_1 (v - d_3)] \\ &= \frac{-1}{d_6} [w_1 (d_3 d_5 - d_1 v) + w_3 (d_1 - d_5) + c_2 (v - d_3)] \end{aligned}$$

$$C_1 = H - au \quad C_2 = H - au$$

$$\boxed{X - \text{Sweep}}$$

$$C_3 = u - a$$

$$C_4 = u + a$$

$$C_5 = \frac{1}{2} \vec{V}^2$$

$$C_6 = C_1 C_4 - C_2 C_3 + C_3 C_5 - C_4 C_5 - C_1 u + C_2 u,$$

$$C_7 = w_4 - w_3 - r + r^2 w_1,$$

$$\begin{aligned} a_1 &= \frac{-1}{C_6} [C_2 w_2 - C_3 w_4 - C_4 w_4 + u w_4 + (a r^2 w_1 + C_4 C_5 w_1 - C_2 u w_1 + C_4 r^2 w_3 - u r^2 w_3)] \\ &= \frac{-1}{C_6} [w_1 (C_4 C_5 - C_2 u) + w_2 (C_2 - C_5) + (C_4 + u) w_4 - r w_3 (C_4 + u) + r^2 w_1 (-C_4 + u)] \\ &= \frac{-1}{C_6} [w_1 (C_4 C_5 - C_2 u) + w_2 (C_2 - C_5) + C_4 (u - C_4)] \end{aligned}$$

$$\begin{aligned} a_2 &= \frac{-1}{C_6} [C_1 w_2 - C_2 w_2 - C_3 w_4 + C_4 w_4 - C_3 r^2 w_1 + C_4 r^2 w_1 - C_1 C_4 w_1 + C_2 C_3 w_1 + C_3 r^2 w_3 - C_4 r^2 w_3] \\ &= \frac{-1}{C_6} [C_2 C_3 w_1 - C_1 C_4 w_1 + C_1 w_2 - C_2 w_2 - C_3 w_4 + C_4 w_4 + C_3 r^2 w_3 - C_4 r^2 w_3 - C_3 r^2 w_1 + C_4 r^2 w_1] \\ &= \frac{-1}{C_6} [w_1 (C_2 C_3 - C_1 C_4) + w_2 (C_1 - C_2) + C_2 (C_4 - C_3)] \end{aligned}$$

$$a_3 = w_3 - r w_1,$$

$$\begin{aligned} a_4 &= \frac{1}{C_6} [C_1 w_2 - C_3 w_4 - C_5 w_2 + u w_4 - C_3 r^2 w_1 + u r^2 w_1 + C_3 C_5 w_1 - C_1 u w_1 + C_3 r^2 w_3 - u r^2 w_3] \\ &= \frac{1}{C_6} [C_3 C_5 w_1 - C_1 u w_1 + C_1 w_2 - C_5 w_2 - C_3 w_4 + u w_4 + C_3 r^2 w_3 - u r^2 w_3 - C_3 r^2 w_1 + u r^2 w_1] \\ &= \frac{1}{C_6} [w_1 (C_3 C_5 - C_1 u) + w_2 (C_1 - C_5) + C_2 (u - C_3)] \end{aligned}$$